

# **Notes on Lance Taylor (2009) *Reconstructing Macroeconomics***

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### **Chapter 1 Social Accounts and Social Relations**

#### *I. SOCIAL ACCOUNTING MATRIX*

Workers receive their incomes from:  $Y_w = wbX + iM_w + Q_w$

What workers spend their incomes on:  $Y_w = PC_w + T_w + S_w$

Workers receive incomes from wages paid to them ( $wbX$ ), interest (at the rate  $i$ ) received on workers' bank deposits ( $M_w$ ), and transfers from the government ( $Q_w$ )

Workers spend their incomes on consumption ( $PC_w$ ), and payment of taxes by workers ( $T_w$ ), and the rest of their incomes are saved ( $S_w$ )

Note:  $w$  is wages per worker;  $X$  is output;  $b$  is the Number of workers ( $L$ ) per unit of output ( $X$ ) or  $b = L/X$ .  $wbX$  is incomes in the form of wages received by all workers combined.

Rentiers receive their incomes from:  $Y_r = \delta P_v V + iM_r$

What rentiers spend their incomes on:  $Y_r = PC_r + T_r + S_r$

$V$  is the volume of equity issued by business firms, all of which (we assume) are held by rentiers,  $P_v$  is the price of equity,  $\delta$  is dividend payout rate.

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Assume you hold 50 shares of a company, with each share selling at a price (average for, say, a year) of Rs.1000. The value of your shares (which is your wealth) is Rs.50,000. If shareholders of this company receive a dividend of, on average, 6%, your annual income from your shares will be Rs. Rs.3000.

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Businesses receive their incomes from:  $Y_b = \Pi PX$

What businesses spend their incomes on:

$$Y_b = \delta P_v V + i L_b + i^* e Z_b^* + T_b + S_r$$

Note:  $\Pi$  is profit share of output,  $P$  is prices (or the price deflator);  $X$  is real output (tonnes of steel);  $PX$  is output in nominal values (Rupees of dollars worth of output).

For firms, the finance for running their businesses comes from the equity raised from the market, loans from domestic banks and foreign financial institutions, and their retained earnings (savings used for future business operations or expansion).

The profit incomes the firms receive are deployed to pay:

Dividends for equity holders (of the shares issued by the firms), interest on the loans they have taken from domestic and foreign sources, and taxes. The rest of the profits are saved.

Governments receive their incomes from:  $Y_g = T_x + T_w + T_r + T_b$

What governments spend their incomes on

$$Y_g = PG + Q_w + iL_g + i^* e Z_g^* + S_g$$

The domestic interest rate is 'i' and the foreign interest rate is  $i^*$

We are not making a distinction between lending rates and deposit rates at this stage.

$L_b$  and  $L_g$  are the loans given by the domestic banking system to businesses and the government, respectively

$Z_b^*$  and  $Z_g^*$  are the foreign loans received by businesses and the government, respectively

We consider here only the tax revenues received by the government.

$T_x$  is taxes minus subsidies on production (indirect taxes).

$T_w$ ,  $T_r$ , and  $T_b$  are taxes received from workers, rentiers and businesses.

Where 'foreign' (all external entities that are engaged in economic transactions with the domestic economy)

receive their incomes from:  $Y_f = e a X + i^* e Z_g^* + i^* e Z_b^*$

What foreign spend their incomes on  $Y_f = PE + i^* e R^* + S_f$

Foreigners receive income from the payments we (the home country) make for our imports ( $eaX$ ) and from the interest paid by businesses and government (in the home country) on the foreign loans they took.

Foreigners spend the income they receive (from us) to pay for the goods and services we export to them ( $PE$ ) and also to pay interest on the foreign exchange reserves that are held by us (the home country) (this is:  $i^* e R^*$  )

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Note: Our foreign exchange reserves are, in effect, the loans we have given to the foreign country.

$a$  is imports (or inputs sourced from abroad) as a share of output ( $X$ );  $e$  is the exchange rate.  $aX$  is the value of imports in, say, dollars, and  $eaX$  is the value of imports in Rupees.

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$$PX = PC_w + PC_r + PG + PE + PI$$

$X$  is “real” output – the aggregate supply of goods and services available for domestic use.  $X$  includes domestic production as well as imports.

GDP is Output minus inputs (here, we consider inputs that are imported),  
So GDP is  $PX$  minus the value of imports ( $eaX$ )

Therefore GDP as expenditures:  $GDP = P(C_w + C_r + G + E + I) - eaX$

GDP as incomes:  $GDP = w_bX + \Pi PX + T_x$

The above is GDP at market prices.

where GDP at factor cost =  $w_bX + \Pi PX$

Note: GDP at factor cost = enough to make payments to factors of production (wages to labour and profits to owners of capital).

GDP at market prices is GDP at factor cost plus  $T_x$

( $T_x$  is indirect taxes (or taxes on production) minus subsidies on production.)

## *II. STOCK-FLOW RELATIONSHIP*

### *Workers*

We assume that all workers' wealth ( $\Omega_w$ ) is held in the form of bank deposits (or currency) ( $M_w$ ).

Stock:  $\Omega_w = M_w$

The workers's savings in a particular year (flow) equals the addition to the stock of workers' wealth.  $\dot{M}_w$  is the addition to the stock of bank deposits.

Therefore, Flow:  $S_w = \dot{M}_w$

The convention is that all sources of funds (for instance, savings) are given a positive sign, and all uses of funds are given a negative sign.

Therefore,  $S_w + (-\dot{M}_w) = 0$

### *Rentiers*

The rentier households hold their wealth (or accumulated savings) in the form of equity ( $P_v V$ ), and bank deposits or currency and ( $M_r$ )

Stock relation:  $\Omega_r = P_v V + M_r$

So their savings this year are an addition to their bank deposits ( $\dot{M}_r$ ) and the new equities they purchase (amounting to  $\dot{V}$ , with a value of  $P_v \dot{V}$ )

Flow relation:  $S_r = P_v \dot{V} + \dot{M}_r$

### *Businesses*

The business's assets are its capital stock (machinery and buildings) as valued by the financial market. The value of the capital stock at replacement value is  $PK$  ( $K$  is capital stock,  $P$  is its price)

$qPK$  is 'asset value' or the valuation put on the capital stock by the financial markets.

On the liability side of business are: domestic and foreign loans, raised equity, and business networth ( $\Omega_b$ ).

Stock relation:  $qpk = L_b + eZ_b^* + P_v V + \Omega_b$

Private investment (PI) (or the addition to capital stock) is financed through fresh loans (both domestic and foreign) and fresh equity raised by

business firms. It is also financed through the firms' own savings (retained earnings).

Therefore:

$$\text{Flow relation: } \dot{PI} = S_b + P_v \dot{V} + \dot{L}_b + e \dot{Z}^*_b$$

Note: New loans ( $\dot{L}_b + e \dot{Z}^*_b$ ) and fresh equity ( $\dot{V}$ ;  $P_v \dot{V}$  is the value of this equity)

*Banks:*

On the asset side of the banking system: the loans given by the banks or the credit issued) to business and government as well as the foreign exchange reserves.

*Assets and Liabilities: Central Bank*

| <i>Liabilities</i>                                                                                             | <i>Assets</i>                              |
|----------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| 1. Currency with the public ( $C^P$ )                                                                          | 1. Credit to the government                |
| 2. Reserves:<br><br>Cash with commercial banks ( $C^B$ )<br>& Commercial banks' deposits with the Central Bank | 2. International reserves (foreign assets) |

The above does not take into account short-term or temporary credit from the Central Bank to commercial banks.

### *Assets and Liabilities: Commercial Banks*

| <i>Liabilities</i>                | <i>Assets</i>                                                                                               |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------|
| 1. Deposits with commercial banks | 1. Credit (or loans) to the private sector and the government                                               |
|                                   | 2. Reserves:<br><br>Cash with commercial banks ( $C^B$ ) & Commercial banks' deposits with the Central Bank |

### *Assets and Liabilities: Banking System (Central and Commercial Banks combined)*

| <i>Liabilities</i>                    | <i>Assets</i>                                                         |
|---------------------------------------|-----------------------------------------------------------------------|
| 1. Currency with the public ( $C^P$ ) | 1. Credit (or loans) to the private sector and the government ( $L$ ) |
| 2. Deposits ( $D$ )                   | 2. International reserves (foreign assets)                            |

[Reserves appear on the asset side of the commercial bank's balance sheet and on the liability side of the central bank's balance sheet.]

We know that:

$C^P + \text{Deposits} = \text{Money supply in the economy (M)}$

Therefore,

Stock relation:  $M = L + eR^*$

Against new loans issued and reserves acquired by the banking system ( $\dot{L} + e\dot{R}^*$ ), new money is created.

$$\text{Flow relation: } S_l + \dot{M} = \dot{L} + e\dot{R}^*$$

$S_l$  represents any saving acquired by the banking system. We will assume that  $S_l = 0$ .

### *Government*

The government's asset is the "full faith and credit"  $\Delta$  behind its debt obligations,  $L_g + eZ^*$

$$\text{Stock relation: } L_g + eZ^* = \Delta$$

The government finances its fiscal deficit (excess of expenditures over debt or negative savings) with fresh loans from domestic and foreign sources.

$$\text{Flow relation: } S_g = \dot{L}_g + e\dot{Z}_g^*$$

### *Foreign*

For foreign entities having economic engagements with "our" domestic economy: the loans we have taken from them ( $Z^*$  in dollars,  $eZ^*$  in Rupees) are their assets. On the other side of the stock relation are the international reserves held by our banking system (which is effectively credit to foreign) and foreign wealth (on account of their relation with our economy or our external debt (foreign loans) minus international reserves )

$$\text{Stock relation: } eR^* + \Omega_f = eZ^*$$

The flow relation:  $S_f + e\dot{R}^* = e\dot{Z}^*$

*The Stock relationships: Assets and Liabilities of various social classes*

| Social classes | Liabilities                       | Assets        |
|----------------|-----------------------------------|---------------|
| 1. Workers     | $\Omega_w$                        | $M_w$         |
| 2. Rentiers    | $\Omega_r$                        | $M_r + P_v V$ |
| 3. Business    | $L_b + eZ_b^* + P_v V + \Omega_b$ | $qPK$         |
| 4. Banks       | $M$                               | $L + eR^*$    |
| 5. Government  | $L_g + eZ^*$                      | $\triangle$   |
| 6. Foreign     | $eR^* + \Omega_f$                 | $eZ^*$        |

Summing across all balance sheets:

$$\triangle + qPK = \Omega_w + \Omega_r + \Omega_b + \Omega_f$$

The “primary wealth” is made up of the government’s debt plus the capital stock valued at its asset price. The primary wealth constitutes the net worths of workers, rentiers, businesses, and the rest of the world.

Note:  $M = M_w + M_r$

$Z = Z_b + Z_g$  and  $L = L_b + L_g$

*The flow relationships: Net increases in Assets and Liabilities of various social classes*

| Social classes | Net Increase in Liabilities                    | Net increase in Assets        |
|----------------|------------------------------------------------|-------------------------------|
| 1. Workers     | $S_w$                                          | $\dot{M}_w$                   |
| 2. Rentiers    | $S_r$                                          | $P_v \dot{V} + \dot{M}_r$     |
| 3. Business    | $S_b + P_v \dot{V} + \dot{L}_b + e\dot{Z}^*_b$ | PI                            |
| 4. Banks       | $S_l + \dot{M}$                                | $\dot{L} + e\dot{R}^*$        |
| 5. Government  | $(S_g)$                                        | $-(\dot{L}_g + e\dot{Z}^*_g)$ |
| 6. Foreign     | $S_f + e\dot{R}^*$                             | $e\dot{Z}^*$                  |

$$\dot{M}_w + \dot{M}_r = \dot{M}$$

$$\dot{Z}^*_g + \dot{Z}^*_b = \dot{Z}^*$$

$$\dot{L}_b + \dot{L}_g = \dot{L}$$

$S_g$  is typically negative = fiscal deficit

Adding up the sources and uses of funds and cancelling out items

$$S_w + S_r + S_b + S_l + S_g + S_f = \text{PI}$$

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## A SIMPLE STOCK-FLOW RELATIONSHIP

For an easier understanding of the relationships, let us consider households, which are workers and rentiers combined.

Let us consider a closed economy - So no foreign in this analysis.

Let us also assume that banks and businesses do not save. So ~~no~~

$$S_b = 0 \quad S_l = 0$$

Households receive incomes from:

$$Y_H = \text{Wages (WBX)} + \text{Transfers (Q}_H) + \text{Returns on financial assets (iM}_H^H + SP_V V)$$

$M_H$  = Currency and deposits held by households = Money Supply in the economy OR  $M$

They spend their incomes on

$$Y_H = \text{Consumption (PC)} + \text{Taxes (T}_H) + \text{Savings (S}_H)$$

All consumption in the economy by households. So  $C_H = C$   
 (Also all currency and deposits with households  
 So  $M_H = M$ )

Businesses receive incomes from

$$Y_b = \text{Profits } (\pi PX)$$

They spend their incomes on

$$Y_b = \text{Payment of dividend on equity and interest on loans } (\delta P_v V + i L_b) + \text{Taxes } (T_b)$$

Governments receive incomes

$$Y_g = \text{Taxes on production (or indirect taxes, } T_x) + \text{Taxes on households (or income taxes, } T_H) + \text{Taxes on business } (T_b)$$

They spend their incomes on

$$Y_g = \text{Govt consumption and investment expenditure } (PGI) + \text{Govt Transfers } (Q_H) + \text{Interest payments } (i L_g) + \text{Government saving } (S_g)$$

[Government savings is typically negative = Fiscal deficit]

We can summarize the income flows (inflows- how incomes are received) and outflows- how they are spent of the main agents in the economy.

|            | INFLows                                                        | OUTFLows                                      |
|------------|----------------------------------------------------------------|-----------------------------------------------|
| Households | $WbX + Q_H + iM_H + SP_V$<br>(wages) Transfers (Returns-asset) | $PC + T_H + S_H$<br>Consumption Taxes Savings |
| Business   | $\pi PX$                                                       | $SP_V + iL_b + T_b$<br>Payments Taxes         |
| Government | $T_x + T_H + T_b$                                              | $PG + Q_H + iL_g + S_g$                       |

Combining the inflows and outflows of the three agents:

$$\underbrace{WbX + \pi PX + T_x}_{\text{GDP at market prices}} = \underbrace{PC + PG + S_H + S_g}_{\text{Savings}}$$

GDP at market prices

Savings



PI

$$\text{OR } \text{GDP at market prices} = P(C + I + G)$$

The translation of savings into investment through the financial system (banks, stock market)

In the above, also note that:

$$i M = i L_g + i L_b$$

(All the money = Currency and deposits in the hands of households; So  $M_H = M$  or money supply).

We know the combined balance sheet of the banking system (commercial banks + Central Bank)

Liabilities  
M (Money supply)

Assets  
L (Loans)  
(In an open economy there will be Reserves as well).

We know that

$i$  on the left side can be seen as the returns received by savers or depositors (of course, no returns on currency).

$i$  on the right side is closer to the lending rate -

A simplified flow relationship shows how savings by households finance investment by business (PI) and fiscal deficit or negative savings by the government.

Household savings  $S_H$  in the form of increased currency or deposits ( $\dot{M}$ ) + increased holdings of equity ( $P_v \dot{V}$ )

These savings reach the banking system. <sup>Banks</sup> They translate the increased money supply into new loans - for the business and government

$$\dot{M} = \dot{L} = \dot{L}_b + \dot{L}_g$$

Businesses use the loans (from banks - which mobilized household savings) and equity (again from households holdings) to finance private investment.

$$P_v \dot{V} + \dot{L}_b = PI$$

Fiscal deficit (which finances government <sup>new</sup> expenditure) financed by banks' ~~new~~ holdings of government securities or  $\dot{L}_g$

## A Simplified Flow Relation

Household Savings :  $S_H = \dot{M} + P_V \dot{V}$

They reach the banks

Banks :  $\dot{M} = \dot{L} (\dot{L}_b + \dot{L}_g)$

The Loans and equity finance

Business :  $\dot{L}_b + P_V \dot{V} = PI$   
investment

Fiscal deficit :  $(S_g) = -\dot{L}_g$

Summarizing :

|            |                           |                         |
|------------|---------------------------|-------------------------|
| Households | $S_H$                     | $\dot{M} + P_V \dot{V}$ |
| Banks      | $\dot{M}$                 | $\dot{L}_b + \dot{L}_g$ |
| Business   | $\dot{L}_b + P_V \dot{V}$ | $PI$                    |
| Government | $(S_g)$                   | $-\dot{L}_g$            |

$S_g$  is typically negative (fiscal deficit)

combining, we see

$$S_H + S_g = PI$$